

Day-2

Data Cleaning

- 1) Handle missing values
- 2) drop missing rows/columns
- 3) Impute with

mean / Median $\rightarrow$ for numerical data

Mode $\rightarrow$ for categorical data

Adv $\rightarrow$ LR, KNN, interpolation

- 4) Remove duplicates
- 5) Handle inconsistencies
- 6) Detect & handle outliers
- 7) fix logic & domain errors

Data Preprocessing

1) Encoding Categorical variables

- Label Encoding (Ordinal):

→ good for ordered categories
like 'Low', 'Medium', 'high'

- One - Hot Encoding (Nominal):

→ For non-ordered categories

like Region

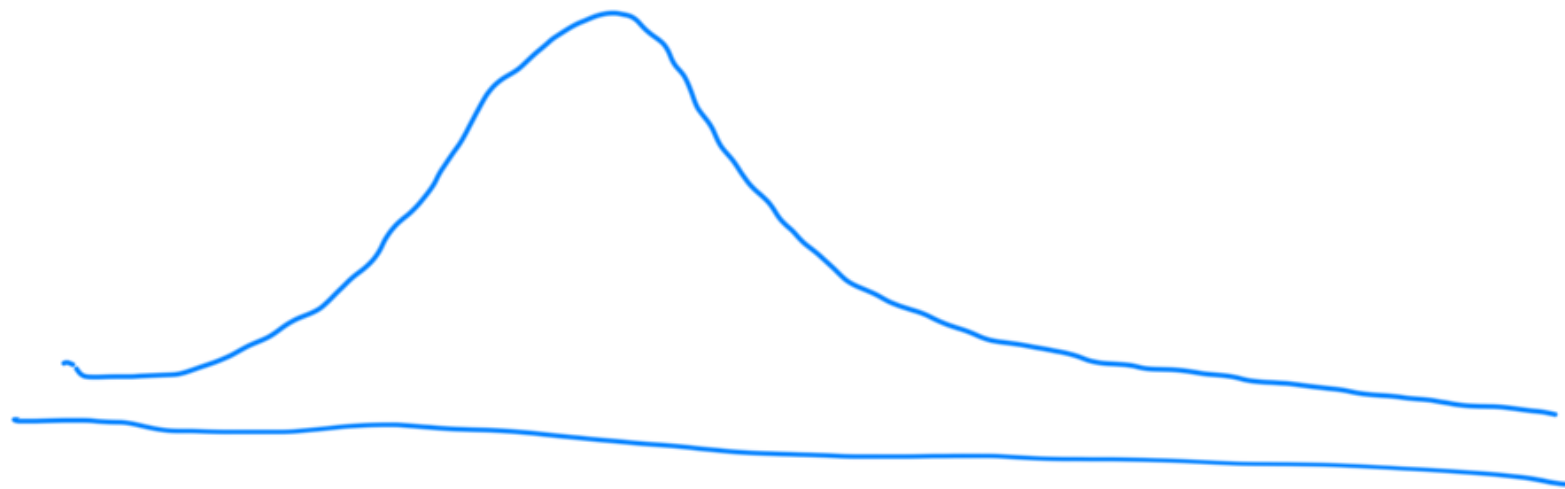
Region | 6 | Salary | Region North

neg			0

2)

Feature Transformation

→ used to handle skewed data, like right or left skewed data



3) Feature Scaling

→ Normalization or Standardization

→ Bringing numerical values to the same scale especially for distance based learning

Age	Salary
23	36,000
24	46,000
25	1,60,000

Age	Salary
0.23	0.36
0.24	0.46
0.25	0.16

→ Normalization (Min-Max Scaling)

→ Scale values b/w 0 and 1.

→ Standardization



Feature Engineering

features

→ mathematical combinations

→ Target based combinations

→ Binning

→ Time-based features

Feature Selection

- Selecting the most imp features & removing the rest,

→ reduce noise & overfitting

✓ reduce model complexity

→ speed up training

→ improves model accuracy

→ make model interpretation easier.